Project Costing Report — Campus Ride-Pooling and Split-Fare System

Course: CS455 - Software Engineering, IIT Kanpur
Repository: CS455-Carpooling/Campus_Ridepooling_Automation
Production URL: https://campus-ridepooling-automation.onrender.com/
Costing Basis: Verified Operating-Cost Analysis & Market Rate Assessment
Currency: Indian Rupees (INR) & United States Dollars (USD)
Exchange Rate Reference: 1 USD = ₹94.00 INR

1. Components in Our Website and Is It Costable

Component Name	Description & Functional Scope	Is It Costable?	Cost Classification & Category
Next.js Application	Server-side rendering, client page layouts, UI components, and API routes for student/admin workflows.	**No** (Directly)	Hosted within the web service container; does not incur separate compute charges.
Custom Node.js & Socket.IO Server	Real-time websockets handling live ride coordination, private trip chat, and instant push notifications.	**Yes**	Billed under Web Service Compute (requires active, non-sleeping process).
PostgreSQL Database (`crp`)	Stores persistent records: users, rides, seat requests, trip progress, fare shares, settlements, complaints, and audit logs.	**Yes**	Billed under Managed Database Compute, Storage, and Backup add-ons.
Google Gemini API Integration	Powers natural language chatbot query parsing, ride matching recommendations, and advisory complaint analysis.	**Yes**	Billed per token (input/output) beyond the free tier quota.
Google Apps Script Email Relay	Handles verification email dispatch and password resets for `@iitk.ac.in` accounts via MailApp relay.	**No** (Current) / **Yes** (Production)	Free within standard Google account sending quotas; costable if upgraded to dedicated SMTP.
GitHub & GitHub Actions	Stores repository source code, manages deployment webhooks, and runs automated CI linting and test suites.	**No**	Free for public repositories within included GitHub Actions monthly minute allowances.
Docker Build Environment	Packages Next.js, Node.js, and Socket.IO into containerized images for automated deployment.	**No**	Included within the hosting provider's build pipeline quota.
Domain & HTTPS Certificate	Provides public domain access (`*.onrender.com`) and automated SSL/TLS encryption.	**No** (Current) / **Yes** (Production)	Render subdomain is free; custom campus domain (`.in` / `.com`) incurs annual registration fees.

2. Current vs. Production Level Differences

Architectural & Operational Metric	Current Setup (Course Evaluation Demo)	Production Level Setup (Full IITK Campus Deployment)
Target User Base & Capacity	5–10 concurrent evaluator / student test accounts.	1,000–5,000 registered student accounts; 100+ concurrent active Socket.IO connections during peak campus hours.
Web Service Compute Tier	Render Free Tier (512 MB RAM, shared CPU).	Render Starter / Standard Paid Tier (1 GB–2 GB RAM, 1 CPU core).
Web Service Availability & Uptime	Spins down (sleeps) after 15 minutes of inactivity; ~60-second cold start latency; capped at 750 free hours/month.	100% Always-On uptime; zero cold-start delay; dedicated CPU allocation for real-time WebSocket connections.
Database Instance Tier	Render Managed PostgreSQL Free Tier.	Render Managed PostgreSQL Paid Basic/Standard Tier.
Database Storage Limit & Expiry	Fixed 1 GB storage limit; **strict 30-day lifecycle limit** (database expires and deletes automatically).	10 GB–50 GB auto-scaling SSD storage; permanent persistent retention with zero deletion risk.
Database Disaster Recovery & Backups	**No automated backups**, no Point-In-Time-Recovery (PITR), no database snapshots.	Automated daily snapshots, 7 to 30-day backup retention, and WAL-based point-in-time recovery.
Google Gemini API Allocation	Free Tier (`gemini-3.1-flash-lite` / `gemini-2.5-flash-lite` : 15 RPM, 1,500 RPD cap).	Paid Pay-As-You-Go Tier (unlimited daily requests, higher RPM/TPM limits for peak usage).
Transactional Email Delivery	Google Apps Script + MailApp (~100 emails/day standard Gmail daily quota).	Dedicated Transactional Email Service (Resend / AWS SES; 3,000–10,000+ emails/month with delivery logs).
Custom Domain & SSL	Default subdomain (`campus-ridepooling-automation.onrender.com`) with shared Render SSL certificate.	Custom Domain (e.g., `ridepool.iitk.ac.in` or custom `.in` domain) with dedicated custom SSL.

3. Current vs. Production Price Comparison

> **Gemini API Token Rate Reference (`gemini-3.1-flash-lite`):**
>
> - **Input Token Cost:** $0.25 per 1,000,000 tokens (≈ ₹0.0235 per 1,000 tokens)
> - **Output Token Cost:** $1.50 per 1,000,000 tokens (≈ ₹0.1410 per 1,000 tokens)

Cost Component	Pricing Basis / Rates	Current Setup (Monthly Cost)	Production Pilot (~1,000 Students)	Production Scale (~5,000 Students)
Web Service Compute	Render Starter $(7/mo)/Standard($ 25/mo)	₹0 ($0.00)	₹658 ($7.00)	₹2,350 ($25.00)
Managed Database Compute	Render Postgres Basic-1GB ($6/mo)/Standard(20$/mo)	₹0 ($0.00)	₹564 ($6.00)	₹1,880 ($20.00)
Database Storage & Backups	Paid storage @ $0.30/GB/month + automated backups	₹0 ($0.00)	₹141 ($1.50)	₹564 ($6.00)
Google Gemini API	Token usage based on ParseError: KaTeX parse error: Expected 'EOF', got '&' at position 12: 0.25/1M in &1.50/1M out	₹0 ($0.00)	₹799 ($8.50)	₹3,995 ($42.50)
Transactional Email Delivery	Google Apps Script (Free) / Resend / AWS SES	₹0 ($0.00)	₹0 ($0.00)	₹1,410 ($15.00)
Custom Domain Registration	Amortized monthly cost of annual `.in` domain (~₹900/yr)	₹0 ($0.00)	₹75 ($0.80)	₹75 ($0.80)
GitHub Actions CI/CD	2,000 free minutes/month allowance	₹0 ($0.00)	₹0 ($0.00)	₹0 ($0.00)
Subtotal (Excl. Tax)	Net monthly operating expense	**₹0 ($0.00)**	**₹2,237 ($23.80)**	**₹10,274 ($109.30)**
Applicable Taxes (18% GST)	Statutory digital service tax in India	₹0 ($0.00)	₹403 ($4.28)	₹1,849 ($19.67)
Final Total Monthly Cost	Net out-of-pocket invoice cost	**₹0 ($0.00)**	**₹2,640 ($28.08)**	**₹12,123 ($128.97)**

4. Total Cost Calculation Formula

$$\text{Total Monthly Operating Cost } (C_{\text{total}}) = (C_{\text{web}} + C_{\text{db}} + C_{\text{storage}} + C_{\text{gemini}} + C_{\text{email}} + C_{\text{domain}} + C_{\text{ci}}) \times (1 + T_{\text{GST}})$$

Where:

- C_{web}: Monthly fixed instance cost for the hosting web service (Render Compute Plan).
- C_{db}: Monthly fixed compute cost for managed database hosting.
- C_{storage}: Database disk expansion fee calculated as:

$$C_{\text{storage}} = S_{\text{data (GB)}} \times R_{\text{storage (\$0.30/GB)}} + C_{\text{backup}}$$

- C_{gemini}: Variable AI API consumption cost calculated as:

$$C_{\text{gemini}} = \left(\frac{N_{\text{input_tokens}}}{1,000,000} \times R_{\text{input (\$0.25)}} \right) + \left(\frac{N_{\text{output_tokens}}}{1,000,000} \times R_{\text{output (\$1.50)}} \right) - \text{Free Quota Allowance}$$

- C_{email}: Monthly fee for dedicated transactional mail delivery over free limits.
- C_{domain}: Monthly amortized domain name renewal fee $\left(\frac{\text{Annual Domain Price}}{12}\right)$.
- C_{ci}: CI/CD pipeline build execution charges beyond the 2,000 free monthly minutes quota.
- T_{GST}: Applicable Goods and Services Tax rate (18% or 0.18).

5. Final Total Calculations

5.1 Current Course-Demo Setup

- Web Service Compute: ₹0.00
- PostgreSQL Database: ₹0.00
- Gemini API Usage: ₹0.00
- Email Relay: ₹0.00
- Custom Domain: ₹0.00
- **Subtotal:** ₹0.00
- **GST (18%):** ₹0.00
- **Final Current Total: ₹0.00 / month**

5.2 Production Pilot Setup (~1,000 Students)

- **Web Service (Starter Tier):** $7.00 \times ₹94.00 = ₹658.00$
- **PostgreSQL (Basic-1GB Tier):** $6.00 \times ₹94.00 = ₹564.00$
- **Database Storage & Backups (5 GB):** $1.50 \times ₹94.00 = ₹141.00$
- **Gemini API Usage:**
  - *Assumptions:* 20,000 API calls/month (matching recommendations, chatbot queries, complaint triage).
  - *Input Tokens:* $20,000 \text{ calls} \times 500 \text{ tokens} = 10,000,000 \text{ tokens}$.

$$\text{Cost} = \left(\frac{10,000,000}{1,000,000} \right) \times \$0.25 = \$2.50 \quad (₹235.00)$$

  - *Output Tokens:* $20,000 \text{ calls} \times 200 \text{ tokens} = 4,000,000 \text{ tokens}$.

$$\text{Cost} = \left(\frac{4,000,000}{1,000,000} \right) \times \$1.50 = \$6.00 \quad (₹564.00)$$

  - *Gemini Subtotal:* $\$8.50 \times ₹94.00 = ₹799.00$
- **Email Delivery (Resend Free Tier <3k emails/mo):** ₹0.00
- **Custom Domain (.in registration amortized):** $\frac{₹900.00}{12} = ₹75.00$
- **Subtotal:** $₹658 + ₹564 + ₹141 + ₹799 + ₹0 + ₹75 = ₹2,237.00$ ($23.80)
- **GST @ 18%:** $₹2,237.00 \times 0.18 = ₹402.66 \approx ₹403.00$ ($4.28)
- **Final Production Pilot Total: ₹2,640.00 / month** ($28.08 / month)

5.3 Production Full-Scale Setup (~5,000 Students)

- **Web Service (Standard Tier):** $25.00 \times ₹94.00 = ₹2,350.00$
- **PostgreSQL (Standard DB Tier):** $20.00 \times ₹94.00 = ₹1,880.00$
- **Database Storage & Backups (20 GB):** $6.00 \times ₹94.00 = ₹564.00$
- **Gemini API Usage:**
  - *Assumptions:* 100,000 API calls/month.
  - *Input Tokens:* $100,000 \text{ calls} \times 500 \text{ tokens} = 50,000,000 \text{ tokens} = \$12.50 \quad (₹1,175.00)$.
  - *Output Tokens:* $100,000 \text{ calls} \times 200 \text{ tokens} = 20,000,000 \text{ tokens} = \$30.00 \quad (₹2,820.00)$.
  - *Gemini Subtotal:* $\$42.50 \times ₹94.00 = ₹3,995.00$
- **Email Delivery Service (Paid Tier):** $15.00 \times ₹94.00 = ₹1,410.00$
- **Custom Domain:** ₹75.00
- **Subtotal:** $₹2,350 + ₹1,880 + ₹564 + ₹3,995 + ₹1,410 + ₹75 = ₹10,274.00$ ($109.30)
- **GST @ 18%:** $₹10,274.00 \times 0.18 = ₹1,849.32 \approx ₹1,849.00$ ($19.67)
- **Final Production Scale Total: ₹12,123.00 / month** ($128.97 / month)

6. Assumptions and Missing Calculations

- Exchange rate fixed at 1 USD = ₹94.00 INR.
- Statutory Goods and Services Tax (GST) rate calculated at 18% on all cloud hosting and API services.
- Gemini API token rates based on the `gemini-3.1-flash-lite` standard pay-as-you-go pricing model ($0.25/1M , $1.50/1M output tokens).
- Average Gemini request payload modeled at 500 input tokens and 200 output tokens per transaction.
- Current course evaluation setup assumes total web service uptime remains within Render's 750 free monthly instance hours limit.
- Developer salaries, software engineering labor, and UI/UX design costs excluded.
- Local developer workstation depreciation, personal electricity usage, and local internet connection fees excluded.
- Egress network bandwidth fees and data migration transfer costs excluded.
- Payment processing and transaction fees excluded (system only records offline cash/UPI settlements).
- Operational monitoring, log aggregation, and error tracking tools (Sentry/Datadog) excluded.
- Third-party SMS verification service fees excluded.